Classifying Motor Imagery EEG by Empirical Mode Decomposition Based on Spatial-Time-Frequency Joint Analysis Approach

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***Abstract*—A novel spatial-time-frequency approach to classify the different mental task in brain computer interface was presented. A high resolution time-frequency spectral was achieved by using Empirical Mode Decomposition and Hilbert-Huang Transform, and the subject specific spatial-time-frequency joint features were extracted from the restricted spectral of multi-channel EEG recordings. A weighting synthetic classifier was built and used to identify the classes of the imaged motions The test results in four subjects showed that the classification accuracy varied between 77.0% and 95.0%, with an average of 85.9%, which suggested that the present method can achieve a reasonable performance in identifying imaged motions compared with previous methods.**

Keywords- EEG; Brain Computer Interface; Spatial-time-frequency joint analysis; Empirical Mode Decomposition; Hilbert-Huang Transform

I. INTRODUCTION

Brain-computer interface (BCI) provides a new brain output pathway that does not depend on the peripheral nerves and muscles, which implements the direct communication and control between the brain and computer devices [1]. Recently, the motor imagery (MI) EEG based BCI has been well developed in BCI research community. The basis underlying of electrophysiological characteristics during MI is that the unilateral MI causes contralateral preponderant event related synchronization (ERS) and desynchronization (ERD) in the motor cortex [2]. The ERD or ERS is about the energy decrease and increase in the rhythmic components of EEG, respectively. Previous discovery showed that the MI caused ERD and ERS in alpha (8-12Hz) and beta (14-30Hz) bands, as well as motor execution [3]. Many studies have also shown that ERD/ERS events are subject specificity [4].

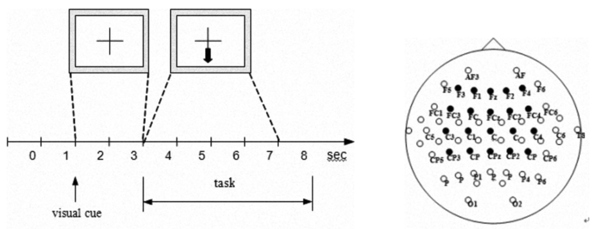
It is well known that EEG signal is highly non-linear and non-stationary, and disturbed by external environmental influences and noise. Due to these problems, how to decode the exact brain mental state from related EEG is the major technical challenge of MI EEG based BCI. Considering the inherent theoretical defects of conventional time-frequency joint analysis methods in studying the non-stationary signal, a new time–frequency method called Empirical Mode Decomposition (EMD) and Hilbert-Huang Transform (HHT) was introduced to analyze the EEG rhythmic components activity. The preliminary results revealed the possible relationship between intrinsic mode functions decomposed by the EMD and the EEG rhythmic components [5].

The objective of the study is to develop new EEG pattern recognition algorithms for EEG based BCI prototype system. In this study, we developed an EMD based spatial-time-frequency joint feature extraction and weighting synthetic classifier, which provides some technical supplements for BCI research and promotes the development and application of this new technique to the clinical.

II. EXPERIMENTAL DATA

The experimental data from BCI2008 competition dataset I were used, which were provided by Berlin BCI group [6]: These data were recorded from 4 healthy subjects. In the whole session MI was performed without feedback. For each subject two classes of MI were selected from the three motion classes: left hand, right hand, and both feet (side chosen by the subject; optionally also both feet). In the first two runs, an arrow was presented as visual cues on a computer screen and could be pointing left, right, or down, respectively. Cues were displayed for a period of 4s during which the subject was instructed to perform the cued motor imagery task. These periods were interleaved with 2s of blank screen and 2s with a fixation cross shown in the center of the screen. The fixation cross was superimposed on the cues, i.e. it was shown for 6s, Fig.1 (a). These data sets are provided with complete marker information.

EEG data were recorded with a sampling frequency of 1 kHz from 59 electrodes placed on the sites corresponding to the International 10/20 system, then filtered with a band-pass filter (0.5 to 200Hz) and down sampled to 100Hz. In this study, EEG data from 20 electrodes indicated in Fig. 1 (b) were chosen. The surface Laplacian was used to reduce spatially smeared due to head volume conductor effect [7].



(a) (b)

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Figure 1. (a) The time course of the experimental paradigm; (b) the 20 electrode locations are used in this study

III. TIME-FREQUENCY ANALYSIS

A. EMD for Non-linear EEG Signal Anlysis

In brief, EMD is a data-driven method to decompose an oscillatory signal into several components with a decreasing order of frequency by using an iterative sifting process. An overview of this method is described as follows:

The goal of the iterative sifting process is to identify frequency modulated oscillations, and decompose signal into a series intrinsic mode functions (IMFs). Each IMF is considered to represent a single oscillation mode in the signal and satisfies two conditions ensuing signal stationary: first, the number of zero crossing and the number of extreme points are equal or differ at most by one; second, the other property is that the mean values of the envelope defined by the local maxima and the mean values of the envelope defined by the local minima are zero at any point. The EMD phase is an adaptive iterative process that applied by shifting their envelope means form the data. More details of the iterative procedure are following that in [8].

A single trail EEG data breaks down into several IMFs and one residue signal after the EMD phase. Form the superposition of the IMFs and residue to reconstruct the data, where $N = 6$ in this case.

$$X(t) = \sum_{n=1}^{N} IMF_i(t) + r_N(t) \tag{1}$$

B. Time-varying Rhythmic Component Description

Each IMF and its Hilbert Transform (HT) form the complex conjugate pair, and an analytic signal can be calculated as [3]:

$$Z_n(t) = IMF_n(t) + j.IMF_n^H(t) = a_n(t)e^{j\theta_n(t)} \tag{2}$$

Where IMF_n^H is the HT of IMF_n, and

$$a_n(t) = \sqrt{IMF_n^2(t) + IMF_n^{H2}(t)} \tag{3}$$

$$\theta_n(t) = \arctan\left(\frac{IMF_n(t)}{IMF_n^H(t)}\right) \tag{4}$$

Where $a_n(t)$ and□$_n(t)$ is the instantaneous features of amplitude and phase. The instantaneous frequency of the IMF_n represents as,

$$\omega_n = \frac{d\theta_n(t)}{dt} \tag{5}$$

Through calculating IMFs of each trail for all subjects, it can be found that the instantaneous frequency ranges of the first four IMFs are empirically close to 4 EEG rhythmic components. Table 1 lists the comparison of frequency ranges between the IMFs and EEG rhythmic bands. Another important fact is, during the MI period, only the power of first two IMFs which cover β and α rhythm respectively show evidently reduction through the grand average approach, and it is also represent a clearly ERD phenomenon.

TABLE I. FREQUENCY RANGES OF IMFS AND EEG RHYTHIM

IMFs	*Frequency Ranges (Hz)*	*EEG Rhythm*	*Frequency Ranges (Hz)*
IMF1	18.3-30.4	β band	14-30
IMF2	8.6-14.5	a band	8-13
IMF3	3.6-7.9	θ band	4-7
IMF4	0.5-3.2	δ band	0.5-3.5

C. The Hilbert-Huang Time-Frequency Spectral

By ignoring the residue, the original EEG data can be reconstructed by the real part of the summation of $Z_n(t)$, and this is also called HHT [8].

$$s(t) = \mathrm{Re}\{\sum_{n=1}^{N} Z_n(t)\} = \mathrm{Re}\{\sum_{n=1}^{N}(a_n(t)e^{j\int \omega_n(t)dt})\} \tag{6}$$

HHT gives both the amplitude and instantaneous frequency of each IMF as function of time. And it can be represented in a three-dimensional plot, in which the amplitude is contoured on the Hilbert-Huang time-frequency (HHTF) spectral as below:

$$H(\omega, t) = \sum_{n=1}^{N} s_n a_n(t) \,|\, e^{j\int \omega_n(t)dt} \,| \tag{7}$$

Where s_n is a switch function and equals 1 when $\omega = \omega_n(t)$, and 0 when else. Fig.2 shows the average HHTF spectral of EEG data from one subject. Contralateral ERD in α rhythm band could be clearly reflected on the HHTF

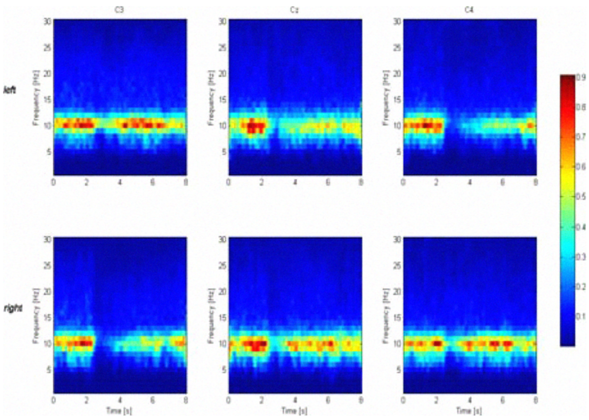

Figure 2. HHTF spectral of EEG data from one subject who performed left or right hand motor imagery. The data was recorded form C3,C4 and Cz electrode channel

IV. Feature Extraction and Classification

A. Time-Frequency Feature and Feature Space Reduction

The first step of the procedure was to extract the time-frequency features from the HHTF spectral. In the time domain, the event-related EEG activity was time-locked in the pre-window lasted 4s. In the frequency domain, the frequency components from 6 Hz to 30Hz which overlapped the β and μ rhythm band with 0.1Hz interval were chosen. Thus a single trail EEG HHTF included 400×250 feature points and such high dimension features could decrease the generalization capability of the classifier with the limited number of training samples. It was crucial to reduce the feature dimensions for the next classification phase.

In this study, we utilized a 2D weighting function to multiple the HHTF to get a small restricted time-frequency plane. Sliding the 2D weighting function over whole spectral plane could generate a series of restricted time-frequency region. The weighting function was a 2-D Gaussian function described as below; this method could also be considered as a 2D Gaussian "blur" filter:

$$\varphi(t,\omega) = \exp(-((t-t')^2 / A^2 + (\omega-\omega')^2 / B^2)) \qquad (8)$$

Where A and B were the spread of Gaussian weighting function in each time and frequency domain; t 'and □ ' was the centre sample point of Gaussian weighting function in each domain. Here, each restricted region choose as [2Hz×0.5s] with 50% overlapped. And the changing pace of t' wasper 25 samples and □ ' was per 10 samples.

In each restricted region, we calculated the mean power of the time-frequency fraction by Eq. (9) and regarded it as the feature represented this region. After this Gaussian "blur" filter, the feature dimension of the HHTF reduced to 25×16. And the index of new time-frequency plane were set as (i, j), which i represent the temporal power segmentation and j represent the frequency bins.

$$E = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \varphi(t,\omega) HHTF(t,\omega) dt d\omega \qquad (9)$$

B. Spatial feature selection

EEG data were recorded in channels overlying the cortex sensorimotor area (SM1) and supplementary motor area (SMA) from the experiment. Reducing the number of dimensions and choosing only fewer channels, which still retain the enough spatial information during the MI period, is benefit for training a classifier efficiently in next phase. Fig.2 shows the average EEG topography map during the left or right MI task, and the difference between the two classes is clear-cut in the spatial domain. In this study, we utilized a class depended maximum mutual information (MMI) scheme to separate the most discriminable signal from all channels.

The MMI between a signal and the classes can be written as $I(C;X)$, where C represents the class labels and X represents the power form a single EEG channel, and the value of MMI can be calculated as [9]:

$$I(C;X) = \sum_{c\in C} \int_{x\in X} p(c,x) \log \frac{p(c,x)}{p(c)p(x)} dx \qquad (10)$$

The right side of Fig.3 shows the MMI topography map from all channels, the regions around C3 and C4 contain the abundant information to discriminate different classes. Thus, we set a selection threshold *Th*. If the MMI value was below than *Th*, the channel did not be selected. Here, we set *Th* = 0.4 by the experience.

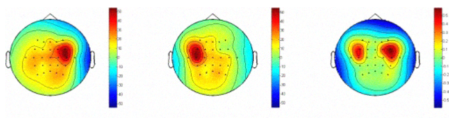

Figure 3. Left: image left hand movement; Middle: image right hand movment; Right: MMI topography map

C. Spatial-Time-Frequenc Joint Fearure Extraction

1) Temporal power oscillation pattern: In nth channel, the time-frequency feature was selected by HHTF and 2D Gaussian filter. By consisting temporal power segmentation of frequency j on the t-f plane to a vector, we obtained the local temporal power oscillation pattern as in Eq. (11),where i_m was the length of temporal segmentation :

$$\mathbf{P}(j,n) = [E(1,j), \ldots E(i,j), \ldots E(i_m,j)] \qquad (11)$$

2) Spatial-time-frequency joint feature: Extracting the temporal power oscillation pattern in each frequency j from the channels selected by MMI scheme and combining them together, we obtained the spatial-time-frequency joint feature as in Eq. (12), where N_m is the number of selected channels and J_m was the largest frequency bins.

$$\mathbf{P} = \begin{pmatrix} \mathbf{p}(1,1) & \cdots & \mathbf{p}(1,n) & \cdots & \mathbf{p}(1,N_m) \\ \vdots & \ddots & & & \vdots \\ \mathbf{p}(j,1) & & \mathbf{p}(j,n) & & \mathbf{p}(j,N_m) \\ \vdots & & & \ddots & \vdots \\ \mathbf{p}(J_m,1) & \cdots & \mathbf{p}(J_m,n) & \cdots & \mathbf{p}(J_m,N_m) \end{pmatrix} \qquad (12)$$

D. Classification

We used each temporal power oscillation pattern $\mathbf{P}(j,n)$ from the training dataset to train a local Support Vector Machine (SVM) classifier independently. In SVM algorithm, the estimation of theoretical error could be calculated by the number of the support vectors (*SV*) and the training samples *l*. The theoretical error corresponds to the classification error rate of the training set. In each local Classifier, we considered the theoretical error as a normalized weighting coefficient as below:

$$W(j,n)=\begin{cases}[1-\frac{E(SV)}{l}]^m, & \frac{E(SV)}{l}\geq T\\ 0, & \frac{E(SV)}{l}<T\end{cases} \quad (13)$$

Where m = 2 and T = 0.6 in this study. Go through all frequency and channels, synthetic decision was made by the following synthetic assignment function:

$$f(x)=\mathrm{sgn}[\sum_{j=1}^{J_m}\sum_{n=1}^{C_m}W(j,n)f_{j,n}(x)] \quad (14)$$

Where $f_{j,n}(x)$ was local decision function in frequency j and channel c. The final decision was made by several local temporal power oscillation patterns which have the strong generalization capability.

V. RESLUTS

The present EMD based spatial-time-frequency synthetic weighting (EMD STF) method was applied to an EEG dataset of four human subjects. The highest offline test classification accuracy was 95.0% for the subject 4 and the mean was 85.9%. The result indicated that the new method described the mental state information comprehensively, and the test classification accuracy was increase with this approach comparing to with the algorithm named time-frequency weighting (TFW) mentioned in Wang's paper [10]. The more results in comparison of the two methods were described in Table 2.

TABLE II. THE CLASSIFICATION ACCURACY OF EMD STF AND TFW

Subject	Classification Accuracy (%)	
	EMD STF	TFW
S1(*left/feet*)	86.0	81.0
S2(*left/right*)	77.0	75.0
S3(*left/feet*)	85.5	82.0
S4(*left/right*)	95.0	93.5
Avg	85.9	82.9

However, the classification accuracy was heavily subject-depended. Comparing the subject 2 to the subject 4, both them performed the left or right hand MI task. But the spatial topography map of subject 4 showed the clearly contralateral control phenomenon, the difference of event-related EEG feature between the two tasks was significant. The accuracy retained to higher than 90% with any pattern recognition algorithm. On the contrary, subject 2 performed the worst classification capability which was less than 80%. On the MI period, bilateral SMA areas showed the large ERD, and the ERD degree and time interval were similar for both subjects. Thus, BCI could identify whether the subject was performing the MI task. But it was hard to distinguish the movement type just depended on the pattern recognition method. This problem could explain as the independence learning process between human and machine. The increase of the MI classification accuracy also relied on the human feedback session, which can change the activate mode of the related motor cortex during MI by utilizing the plasticity of brain.

In conclusion, this paper presented a novel approach to classify the different MI related EEG data. EMD-HHT method could give a high resolution time-frequency spectral, and the spatial-time-frequency joint features were extracted from the HHTFs recorded on multi channels. The selection of channels based on MMI scheme. Weighting synthetic classifier was used to identify the imaged movement type. The test classification accuracy was better than pervious study in [10]. The future study will focus on developing the on-line feedback session and test our algorithm in the real environment.

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